



Supplementary Information for

A structure-preserving neural density functional for the ions of a polymer electrolyte

Liyao Lyu

Liyao Lyu.

E-mail: lyuliyao@ustc.edu.cn

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S1. Properties that hold by construction

The ions are described by their number densities $n_\alpha(\mathbf{r})$, $\alpha \in \{+, -\}$, collected in $\mathbf{n} = (n_+, n_-)$, with valences $z_\pm = \pm 1$. Energies are in units of $k_B T$ and charges in units of e . The free energy of the main text is

$$F[\mathbf{n}] = F_{\text{id}}[\mathbf{n}] + F_{\text{Coul}}[\mathbf{n}] + F_{\text{ex}}^\theta[\mathbf{n}], \quad F_{\text{id}} = \sum_\alpha \int d^3 r n_\alpha [\ln(n_\alpha \Lambda^3) - 1], \quad F_{\text{Coul}} = \frac{l_B}{2} \iint d^3 r d^3 r' \frac{\rho_Z(\mathbf{r}) \rho_Z(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|}, \quad [\text{S1}]$$

with $\rho_Z = \sum_\alpha z_\alpha n_\alpha$ the charge density in units of e , and the neural functional is

$$F_{\text{ex}}^\theta[\mathbf{n}] = \frac{1}{2} \sum_{\alpha\beta} \iint d^3 r d^3 r' n_\alpha(\mathbf{r}) u_{\alpha\beta}(|\mathbf{r} - \mathbf{r}'|) n_\beta(\mathbf{r}') + \int d^3 r \Phi_\theta(\mathbf{h}(\mathbf{r})), \quad h_{m\alpha} = K_m \star n_\alpha, \quad [\text{S2}]$$

with symmetric pair kernels $u_{\alpha\beta} = u_{\beta\alpha} = \sum_m a_{\alpha\beta m} K_m$. The radial kernels have the form $K_m(r) = B_m(r)[1 + g_m(r)]$, where $B_m(r) = (2\pi s_m^2)^{-3/2} \exp[-r^2/(2s_m^2)]$ are normalized Gaussians with six widths from 0.25σ to 1σ , and g is a smooth learned function of r/σ . For bounded g_m , the Gaussian envelope ensures that $K_m(r) \rightarrow 0$ as $r \rightarrow \infty$. The same decay holds for the at-most-linear growth of the softplus networks used here. The kernels K_m and $u_{\alpha\beta}$ are therefore integrable with finite radial moments. Their Fourier transforms are real, even, smooth functions of k , with $\tilde{K}_m(k) = \tilde{K}_m(0) + O(k^2)$ and $\tilde{u}_{\alpha\beta}(k) = \tilde{u}_{\alpha\beta}(0) + O(k^2)$.

For numerical evaluation, the radial Fourier integral is truncated at 5σ and computed by midpoint quadrature with spacing 0.01σ :

$$\tilde{K}_m(k) = 4\pi \int_0^\infty r^2 K_m(r) \frac{\sin(kr)}{kr} dr \simeq 4\pi \int_0^{5\sigma} r^2 K_m(r) \frac{\sin(kr)}{kr} dr,$$

where $\sin(kr)/(kr)$ is evaluated by its limiting value 1 at $k = 0$. In planar geometry, the weighted densities are evaluated as products in Fourier space, $\tilde{h}_{m\alpha}(k) = \tilde{K}_m(k) \tilde{n}_\alpha(k)$.

The pointwise map Φ_θ is a multilayer perceptron with smooth softplus activations and a linear output, so it is twice continuously differentiable and has no explicit dependence on position or direction. As in the main text, θ denotes all learned parameters: the coefficients $a_{\alpha\beta m}$ and the weights of Φ_θ and g . We write $\mu_\alpha^{\text{int}} = \delta(F_{\text{Coul}} + F_{\text{ex}}^\theta)/\delta n_\alpha$ for the interaction chemical potential and $\mathbf{f}_\alpha^{\text{int}} = -n_\alpha \nabla \mu_\alpha^{\text{int}}$ for the internal force density.

With $\Phi_{m\alpha} = \partial \Phi_\theta / \partial h_{m\alpha}$ and $\Phi_{m\alpha, m'\beta} = \partial^2 \Phi_\theta / \partial h_{m\alpha} \partial h_{m'\beta}$ evaluated at $\mathbf{h}(\mathbf{r})$, and since a radial kernel is its own adjoint under convolution, the first and second derivatives of the many-body term are

$$\frac{\delta}{\delta n_\alpha(\mathbf{r})} \int \Phi_\theta(\mathbf{h}) = \sum_m (K_m \star \Phi_{m\alpha})(\mathbf{r}), \quad \frac{\delta^2}{\delta n_\alpha(\mathbf{r}) \delta n_\beta(\mathbf{r}')} \int \Phi_\theta(\mathbf{h}) = \sum_{mm'} \int d^3 r'' K_m(|\mathbf{r} - \mathbf{r}''|) \Phi_{m\alpha, m'\beta}(\mathbf{r}'') K_{m'}(|\mathbf{r}'' - \mathbf{r}'|). \quad [\text{S3}]$$

At uniform density the Hessian $H_{m\alpha, m'\beta} = \Phi_{m\alpha, m'\beta}$ is a constant symmetric matrix, and the bulk kernel of the excess functional is

$$W_{\alpha\beta}(k) = \tilde{u}_{\alpha\beta}(k) + \sum_{mm'} \tilde{K}_m(k) H_{m\alpha, m'\beta} \tilde{K}_{m'}(k). \quad [\text{S4}]$$

Proposition 1 (translation and rotation covariance). Let $g = (R, \mathbf{a})$ with $R \in O(3)$ act on points by $g\mathbf{r} = R\mathbf{r} + \mathbf{a}$ and on densities by $(T_g \mathbf{n})_\alpha(\mathbf{r}) = n_\alpha(g^{-1}\mathbf{r})$. For every such transformation compatible with the domain and every density field specified above:

- i $F_{\text{Coul}}[T_g \mathbf{n}] + F_{\text{ex}}^\theta[T_g \mathbf{n}] = F_{\text{Coul}}[\mathbf{n}] + F_{\text{ex}}^\theta[\mathbf{n}]$,
- ii $\mu_\alpha^{\text{int}}[T_g \mathbf{n}](g\mathbf{r}) = \mu_\alpha^{\text{int}}[\mathbf{n}](\mathbf{r})$,
- iii $\mathbf{f}_\alpha^{\text{int}}[T_g \mathbf{n}](g\mathbf{r}) = R \mathbf{f}_\alpha^{\text{int}}[\mathbf{n}](\mathbf{r})$.

Proof. The weighted densities are scalar fields: substituting $\mathbf{r} = g\mathbf{x}$, $\mathbf{r}' = g\mathbf{s}$ in

$$\begin{aligned} h_{m\alpha}[T_g \mathbf{n}](\mathbf{r}) &= \int K_m(|\mathbf{r} - \mathbf{r}'|) T_g n_\alpha(\mathbf{r}') d^3 \mathbf{r}' = \int K_m(|\mathbf{r} - \mathbf{r}'|) n_\alpha(g^{-1}\mathbf{r}') d^3 \mathbf{r}' \\ &= \int K_m(|g\mathbf{x} - g\mathbf{s}|) n_\alpha(\mathbf{s}) d^3 \mathbf{s} = \int K_m(|\mathbf{x} - \mathbf{s}|) n_\alpha(\mathbf{s}) d^3 \mathbf{s} = h_{m\alpha}[\mathbf{n}](\mathbf{x}) = h_{m\alpha}[\mathbf{n}](g^{-1}\mathbf{r}), \end{aligned}$$

where $d^3 r = |\det R| d^3 s = d^3 s$. Since Φ_θ has no explicit dependence on position, the covariance of the weighted densities and the substitution $\mathbf{r} = g\mathbf{s}$ give

$$\begin{aligned} \int d^3 r \Phi_\theta(\mathbf{h}[T_g \mathbf{n}](\mathbf{r})) &= \int d^3 r \Phi_\theta(\mathbf{h}[\mathbf{n}](g^{-1}\mathbf{r})) \\ &= \int d^3 s \Phi_\theta(\mathbf{h}[\mathbf{n}](\mathbf{s})). \end{aligned}$$

Thus the many-body term is invariant. The pair term and F_{Coul} are double integrals of radial kernels and are invariant by the same two substitutions. This is (i).

Applying (i) to $\mathbf{n} + \epsilon \boldsymbol{\eta}$ and using the linearity of T_g gives

$$\begin{aligned} F_{\text{Coul}}[T_g \mathbf{n} + \epsilon T_g \boldsymbol{\eta}] + F_{\text{ex}}^\theta[T_g \mathbf{n} + \epsilon T_g \boldsymbol{\eta}] \\ = F_{\text{Coul}}[\mathbf{n} + \epsilon \boldsymbol{\eta}] + F_{\text{ex}}^\theta[\mathbf{n} + \epsilon \boldsymbol{\eta}]. \end{aligned}$$

Differentiating with respect to ϵ at $\epsilon = 0$ yields

$$\sum_{\alpha} \int d^3 r \mu_{\alpha}^{\text{int}}[T_g \mathbf{n}](\mathbf{r}) \eta_{\alpha}(g^{-1} \mathbf{r}) = \sum_{\alpha} \int d^3 s \mu_{\alpha}^{\text{int}}[\mathbf{n}](\mathbf{s}) \eta_{\alpha}(\mathbf{s}).$$

Substituting $\mathbf{r} = g\mathbf{s}$ on the left-hand side and rearranging, we obtain

$$\sum_{\alpha} \int d^3 s \left[\mu_{\alpha}^{\text{int}}[T_g \mathbf{n}](g\mathbf{s}) - \mu_{\alpha}^{\text{int}}[\mathbf{n}](\mathbf{s}) \right] \eta_{\alpha}(\mathbf{s}) = 0.$$

Since this holds for every perturbation $\boldsymbol{\eta}$,

$$\mu_{\alpha}^{\text{int}}[T_g \mathbf{n}](g\mathbf{r}) = \mu_{\alpha}^{\text{int}}[\mathbf{n}](\mathbf{r}),$$

which proves (ii).

By (ii) and the chain rule, $\nabla \mu_{\alpha}^{\text{int}}[T_g \mathbf{n}](\mathbf{r}) = R(\nabla \mu_{\alpha}^{\text{int}}[\mathbf{n}])(g^{-1} \mathbf{r})$. Multiplying by $-(T_g \mathbf{n})_{\alpha}(\mathbf{r})$ gives (iii). $\square$

The proof includes reflections. No equivariant architecture is needed, because the covariance of the force follows from the invariance of one scalar.

Proposition 2 (integrability). For every density field specified above,

$$\frac{\delta \mu_{\alpha}^{\text{int}}(\mathbf{r})}{\delta n_{\beta}(\mathbf{r}')} = \frac{\delta \mu_{\beta}^{\text{int}}(\mathbf{r}')}{\delta n_{\alpha}(\mathbf{r})}, \quad [\text{S5}]$$

so the direct correlation functions satisfy $c_{\alpha\beta}(\mathbf{r}, \mathbf{r}') = c_{\beta\alpha}(\mathbf{r}', \mathbf{r})$, and in the bulk $W_{\alpha\beta}(k) = W_{\beta\alpha}(k)$ is real and depends on $|\mathbf{k}|$ only.

Proof. Since Φ_{θ} is twice continuously differentiable,

$$\Phi_{m\alpha, m'\beta}(\mathbf{r}'') = \Phi_{m'\beta, m\alpha}(\mathbf{r}'').$$

Using Eq. S3, the symmetry of the many-body Hessian follows explicitly:

$$\begin{aligned} & \frac{\delta^2}{\delta n_{\alpha}(\mathbf{r}) \delta n_{\beta}(\mathbf{r}')} \int d^3 s \Phi_{\theta}(\mathbf{h}(\mathbf{s})) \\ &= \sum_{mm'} \int d^3 r'' K_m(|\mathbf{r} - \mathbf{r}''|) \Phi_{m\alpha, m'\beta}(\mathbf{r}'') K_{m'}(|\mathbf{r}'' - \mathbf{r}'|) \\ &= \sum_{mm'} \int d^3 r'' K_{m'}(|\mathbf{r}' - \mathbf{r}''|) \Phi_{m'\beta, m\alpha}(\mathbf{r}'') K_m(|\mathbf{r}'' - \mathbf{r}|) \\ &= \frac{\delta^2}{\delta n_{\beta}(\mathbf{r}') \delta n_{\alpha}(\mathbf{r})} \int d^3 s \Phi_{\theta}(\mathbf{h}(\mathbf{s})), \end{aligned}$$

where the last equality follows by relabelling $m \leftrightarrow m'$.

The pair and Coulomb contributions to the interaction Hessian are also symmetric:

$$u_{\alpha\beta}(|\mathbf{r} - \mathbf{r}'|) = u_{\beta\alpha}(|\mathbf{r}' - \mathbf{r}|), \quad \frac{l_B z_{\alpha} z_{\beta}}{|\mathbf{r} - \mathbf{r}'|} = \frac{l_B z_{\beta} z_{\alpha}}{|\mathbf{r}' - \mathbf{r}|}.$$

Combining these contributions gives

$$\frac{\delta \mu_{\alpha}^{\text{int}}(\mathbf{r})}{\delta n_{\beta}(\mathbf{r}')} = \frac{\delta \mu_{\beta}^{\text{int}}(\mathbf{r}')}{\delta n_{\alpha}(\mathbf{r})}.$$

In a uniform state, $H_{m\alpha, m'\beta} = H_{m'\beta, m\alpha}$, so Eq. S4 gives

$$\begin{aligned} W_{\beta\alpha}(k) &= \tilde{u}_{\beta\alpha}(k) + \sum_{mm'} \tilde{K}_m(k) H_{m\beta, m'\alpha} \tilde{K}_{m'}(k) \\ &= \tilde{u}_{\alpha\beta}(k) + \sum_{mm'} \tilde{K}_{m'}(k) H_{m'\beta, m\alpha} \tilde{K}_m(k) \\ &= \tilde{u}_{\alpha\beta}(k) + \sum_{mm'} \tilde{K}_m(k) H_{m\alpha, m'\beta} \tilde{K}_{m'}(k) \\ &= W_{\alpha\beta}(k). \end{aligned}$$

The Fourier transforms of the real radial kernels are real and depend only on $k = |\mathbf{k}|$; hence the same holds for $W_{\alpha\beta}(k)$. $\square$

Lemma (Helmholtz condition). A family $\psi_\alpha[\mathbf{n}](\mathbf{r})$ on a convex set of densities is a functional gradient, $\psi_\alpha = \delta F / \delta n_\alpha$, if and only if $\delta\psi_\alpha(\mathbf{r}) / \delta n_\beta(\mathbf{r}') = \delta\psi_\beta(\mathbf{r}') / \delta n_\alpha(\mathbf{r})$. In that case, with $\mathbf{n}_\lambda = \mathbf{n}_0 + \lambda(\mathbf{n} - \mathbf{n}_0)$,

$$F[\mathbf{n}] = F[\mathbf{n}_0] + \int_0^1 d\lambda \sum_\alpha \int d^3r \psi_\alpha[\mathbf{n}_\lambda](\mathbf{r}) (n_\alpha - n_{0,\alpha})(\mathbf{r}), \quad [\text{S6}]$$

independent of the path.

Proof. Necessity follows from the symmetry of second derivatives: if $\psi_\alpha = \delta F / \delta n_\alpha$, then

$$\frac{\delta\psi_\alpha(\mathbf{r})}{\delta n_\beta(\mathbf{r}')} = \frac{\delta^2 F}{\delta n_\beta(\mathbf{r}') \delta n_\alpha(\mathbf{r})} = \frac{\delta^2 F}{\delta n_\alpha(\mathbf{r}) \delta n_\beta(\mathbf{r}')} = \frac{\delta\psi_\beta(\mathbf{r}')}{\delta n_\alpha(\mathbf{r})}.$$

For sufficiency, keep $\mathbf{n}_0$ fixed and define F by Eq. S6, with $\mathbf{n}_\lambda = \mathbf{n}_0 + \lambda(\mathbf{n} - \mathbf{n}_0)$. Since

$$\frac{\delta n_{\lambda,\alpha}(\mathbf{r})}{\delta n_\beta(\mathbf{r}')} = \lambda \delta_{\alpha\beta} \delta(\mathbf{r} - \mathbf{r}'),$$

functional differentiation gives

$$\begin{aligned} \frac{\delta F}{\delta n_\beta(\mathbf{r}')} &= \int_0^1 d\lambda \left[\psi_\beta[\mathbf{n}_\lambda](\mathbf{r}') \right. \\ &\quad \left. + \lambda \sum_\alpha \int d^3r \frac{\delta\psi_\alpha[\mathbf{n}](\mathbf{r})}{\delta n_\beta(\mathbf{r}')} \Big|_{\mathbf{n}=\mathbf{n}_\lambda} (n_\alpha - n_{0,\alpha})(\mathbf{r}) \right]. \end{aligned}$$

By the assumed symmetry,

$$\frac{\delta\psi_\alpha[\mathbf{n}](\mathbf{r})}{\delta n_\beta(\mathbf{r}')} \Big|_{\mathbf{n}=\mathbf{n}_\lambda} = \frac{\delta\psi_\beta[\mathbf{n}](\mathbf{r}')}{\delta n_\alpha(\mathbf{r})} \Big|_{\mathbf{n}=\mathbf{n}_\lambda}.$$

Moreover, the chain rule along $\mathbf{n}_\lambda$ gives

$$\frac{\partial}{\partial \lambda} \psi_\beta[\mathbf{n}_\lambda](\mathbf{r}') = \sum_\alpha \int d^3r \frac{\delta\psi_\beta[\mathbf{n}](\mathbf{r}')}{\delta n_\alpha(\mathbf{r})} \Big|_{\mathbf{n}=\mathbf{n}_\lambda} (n_\alpha - n_{0,\alpha})(\mathbf{r}).$$

Combining these identities, we obtain

$$\begin{aligned} \frac{\delta F}{\delta n_\beta(\mathbf{r}')} &= \int_0^1 d\lambda \left[\psi_\beta[\mathbf{n}_\lambda](\mathbf{r}') + \lambda \frac{\partial}{\partial \lambda} \psi_\beta[\mathbf{n}_\lambda](\mathbf{r}') \right] \\ &= \int_0^1 d\lambda \frac{\partial}{\partial \lambda} [\lambda \psi_\beta[\mathbf{n}_\lambda](\mathbf{r}')] \\ &= [\lambda \psi_\beta[\mathbf{n}_\lambda](\mathbf{r}')]_0^1 = \psi_\beta[\mathbf{n}](\mathbf{r}'). \end{aligned}$$

Thus $\psi_\alpha = \delta F / \delta n_\alpha$, and its line integral along any smooth path between two densities equals the difference in F at the endpoints, establishing path independence. $\square$

Eq. S6 is the functional line integration used by neural functionals that learn the one-body direct correlation function $c^{(1)} = -\delta F_{\text{ex}} / \delta n$ directly. Such a network has no reason to satisfy the Helmholtz condition exactly, and the value of Eq. S6 then depends on the path. The scalar form of Eq. 4 satisfies it identically.

Proposition 3 (Noether sum rules and stress tensor). For every density field specified above, in or out of equilibrium:

$$\text{i } \sum_\alpha \int \mathbf{f}_\alpha^{\text{int}} d^3r = 0,$$

$$\text{ii } \text{on } \mathbb{R}^3, \sum_\alpha \int \mathbf{r} \times \mathbf{f}_\alpha^{\text{int}} d^3r = 0,$$

Proof. We use the Einstein summation convention for Cartesian indices.

(i) For a translation by $\mathbf{a}$,

$$(T_{\mathbf{a}}\mathbf{n})_\alpha(\mathbf{r}) = n_\alpha(\mathbf{r} - \mathbf{a}), \quad \frac{\partial (T_{\mathbf{a}}\mathbf{n})_\alpha(\mathbf{r})}{\partial a_p} \Big|_{\mathbf{a}=0} = -\partial_p n_\alpha(\mathbf{r}).$$

Differentiating Proposition 1(i) with respect to a_p gives

$$\begin{aligned}
0 &= \frac{\partial}{\partial a_p} \left(F_{\text{Coul}}[T_{\mathbf{a}} \mathbf{n}] + F_{\text{ex}}^\theta[T_{\mathbf{a}} \mathbf{n}] \right) \Big|_{\mathbf{a}=0} \\
&= - \sum_{\alpha} \int d^3 r \mu_{\alpha}^{\text{int}}(\mathbf{r}) \partial_p n_{\alpha}(\mathbf{r}) \\
&= \sum_{\alpha} \int d^3 r n_{\alpha}(\mathbf{r}) \partial_p \mu_{\alpha}^{\text{int}}(\mathbf{r}) \\
&= - \sum_{\alpha} \int d^3 r f_{\alpha,p}^{\text{int}}(\mathbf{r}).
\end{aligned}$$

Since this holds for each Cartesian component,

$$\sum_{\alpha} \int d^3 r \mathbf{f}_{\alpha}^{\text{int}}(\mathbf{r}) = 0.$$

(ii) On $\mathbb{R}^3$, let g_{ϑ} be a rotation by angle ϑ about the unit vector $\hat{\omega}$. Then

$$\frac{\partial (T_{g_{\vartheta}} \mathbf{n})_{\alpha}(\mathbf{r})}{\partial \vartheta} \Big|_{\vartheta=0} = -(\hat{\omega} \times \mathbf{r}) \cdot \nabla n_{\alpha}(\mathbf{r}).$$

Using rotational invariance and $\nabla \cdot (\hat{\omega} \times \mathbf{r}) = 0$, integration by parts gives

$$\begin{aligned}
0 &= \frac{\partial}{\partial \vartheta} \left(F_{\text{Coul}}[T_{g_{\vartheta}} \mathbf{n}] + F_{\text{ex}}^\theta[T_{g_{\vartheta}} \mathbf{n}] \right) \Big|_{\vartheta=0} \\
&= - \sum_{\alpha} \int d^3 r \mu_{\alpha}^{\text{int}}(\hat{\omega} \times \mathbf{r}) \cdot \nabla n_{\alpha} \\
&= \sum_{\alpha} \int d^3 r n_{\alpha}(\hat{\omega} \times \mathbf{r}) \cdot \nabla \mu_{\alpha}^{\text{int}} \\
&= -\hat{\omega} \cdot \sum_{\alpha} \int d^3 r \mathbf{r} \times \mathbf{f}_{\alpha}^{\text{int}}.
\end{aligned}$$

Since $\hat{\omega}$ is arbitrary,

$$\sum_{\alpha} \int d^3 r \mathbf{r} \times \mathbf{f}_{\alpha}^{\text{int}} = 0.$$

□

The screening sum rules follow from the bulk linear response of the same functional. For a uniform stationary state with densities $\bar{n}_{\alpha}$, write $\bar{N} = \text{diag}(\bar{n}_{+}, \bar{n}_{-})$ and $\mathbf{z} = (1, -1)^{\top}$. Linearizing the Euler–Lagrange equation $\ln n_{\alpha} + \mu_{\alpha}^{\text{int}} + V_{\alpha} = \text{const}_{\alpha}$ and Fourier transforming gives $[\bar{N}^{-1} + (4\pi l_B/k^2)\mathbf{z}\mathbf{z}^{\top} + W(k)]\delta\tilde{\mathbf{n}}(\mathbf{k}) = -\delta\tilde{\mathbf{V}}(\mathbf{k})$ for $k \neq 0$. Thus the response matrix defined by $\delta\tilde{\mathbf{n}} = -S(k)\delta\tilde{\mathbf{V}}$ satisfies

$$S(k)^{-1} = \bar{N}^{-1} + \frac{4\pi l_B}{k^2} \mathbf{z}\mathbf{z}^{\top} + W(k). \quad [\text{S7}]$$

This is the Ornstein–Zernike relation $S^{-1} = \bar{N}^{-1} - c$, with $c = -(4\pi l_B/k^2)\mathbf{z}\mathbf{z}^{\top} - W$. For a stable uniform state, the fluctuation-response relation identifies $S(k)$ as the structure factor, $S_{\alpha\beta}(k) = V^{-1}\langle \delta\tilde{n}_{\alpha}(\mathbf{k}) \delta\tilde{n}_{\beta}(-\mathbf{k}) \rangle$. The explicit Coulomb term and the regular small- k behavior of $W(k)$ yield the following screening identities.

Proposition 4 (perfect-screening sum rules). Let $A(k) = \bar{N}^{-1} + W(k)$ be the short-range part of $S(k)^{-1}$ in Eq. S7. If $A(0)$ is invertible and $s_0 = \mathbf{z}^{\top} A(0)^{-1} \mathbf{z} \neq 0$, then for every learned W

$$\frac{S_{ZZ}(k)}{2\bar{n}} = \frac{k^2}{\kappa_D^2} + O(k^4), \quad \mathbf{a}^{\top} S(k) \mathbf{z} = O(k^2) \text{ for every fixed } \mathbf{a}, \quad [\text{S8}]$$

where $S_{ZZ} = \mathbf{z}^{\top} S \mathbf{z}$ is the charge structure factor, $\bar{n}_{+} = \bar{n}_{-} = \bar{n}$ and $\kappa_D^2 = 4\pi l_B \sum_{\alpha} \bar{n}_{\alpha} z_{\alpha}^2 = 8\pi l_B \bar{n}$.

Proof. By the small- k expansions of the radial kernels and Eq. S4,

$$W(k) = W(0) + O(k^2), \quad A(k) = A(0) + O(k^2).$$

Since $A(0)$ is invertible, $A(k)$ remains invertible for sufficiently small k , with

$$A(k)^{-1} = A(0)^{-1} + O(k^2).$$

Consequently,

$$s(k) \equiv \mathbf{z}^\top A(k)^{-1} \mathbf{z} = s_0 + O(k^2), \quad s_0 = \mathbf{z}^\top A(0)^{-1} \mathbf{z} \neq 0.$$

Equation S7 can be written as

$$S(k)^{-1} = A(k) + \frac{4\pi l_B}{k^2} \mathbf{z} \mathbf{z}^\top.$$

Applying the Sherman–Morrison formula gives

$$\begin{aligned} S(k) &= A(k)^{-1} - \frac{(4\pi l_B/k^2) A(k)^{-1} \mathbf{z} \mathbf{z}^\top A(k)^{-1}}{1 + (4\pi l_B/k^2) s(k)} \\ &= A(k)^{-1} - \frac{A(k)^{-1} \mathbf{z} \mathbf{z}^\top A(k)^{-1}}{s(k) + k^2/(4\pi l_B)}. \end{aligned}$$

The denominator is nonzero for sufficiently small k because it tends to $s_0 \neq 0$.

Contracting with $\mathbf{z}$ on both sides yields

$$\begin{aligned} S_{ZZ}(k) &= \mathbf{z}^\top S(k) \mathbf{z} \\ &= s(k) - \frac{s(k)^2}{s(k) + k^2/(4\pi l_B)} \\ &= \frac{k^2}{4\pi l_B} \frac{s(k)}{s(k) + k^2/(4\pi l_B)} \\ &= \frac{k^2}{4\pi l_B} \left[1 + \frac{k^2}{4\pi l_B s(k)} \right]^{-1}. \end{aligned}$$

Since $s(k) = s_0 + O(k^2)$ and $s_0 \neq 0$,

$$\frac{k^2}{4\pi l_B s(k)} = \frac{k^2}{4\pi l_B s_0} + O(k^4).$$

Expanding the reciprocal therefore gives

$$\begin{aligned} S_{ZZ}(k) &= \frac{k^2}{4\pi l_B} \left[1 - \frac{k^2}{4\pi l_B s_0} + O(k^4) \right] \\ &= \frac{k^2}{4\pi l_B} - \frac{k^4}{(4\pi l_B)^2 s_0} + O(k^6). \end{aligned}$$

Using $\kappa_D^2 = 8\pi l_B \bar{n}$, we obtain

$$\frac{S_{ZZ}(k)}{2\bar{n}} = \frac{k^2}{\kappa_D^2} + O(k^4).$$

For any fixed vector $\mathbf{a}$, the same inverse formula gives

$$\begin{aligned} \mathbf{a}^\top S(k) \mathbf{z} &= \mathbf{a}^\top A(k)^{-1} \mathbf{z} - \frac{[\mathbf{a}^\top A(k)^{-1} \mathbf{z}] s(k)}{s(k) + k^2/(4\pi l_B)} \\ &= \frac{k^2}{4\pi l_B} \frac{\mathbf{a}^\top A(k)^{-1} \mathbf{z}}{s(k) + k^2/(4\pi l_B)}. \end{aligned}$$

Finally,

$$\mathbf{a}^\top A(k)^{-1} \mathbf{z} = \mathbf{a}^\top A(0)^{-1} \mathbf{z} + O(k^2), \quad \frac{1}{s(k) + k^2/(4\pi l_B)} = \frac{1}{s_0} + O(k^2),$$

so

$$\mathbf{a}^\top S(k) \mathbf{z} = \frac{k^2}{4\pi l_B s_0} \mathbf{a}^\top A(0)^{-1} \mathbf{z} + O(k^4) = O(k^2).$$

This proves both screening identities. □

The leading k^2 coefficient is the Stillinger–Lovett second-moment condition (1). It is fixed by the explicit Coulomb term, while short-range correlations enter at order k^4 .

S2. Data, force matching and prediction

S2.1 Model and data. The molecular dynamics model and simulation protocol are described in the main text. The additional interaction parameters are the FENE bond constants $K = 30\epsilon/\sigma^2$ and $R_0 = 1.5\sigma$, the Lennard-Jones mixing rule $\sigma_{ij} = (\sigma_i + \sigma_j)/2$, and a 2σ cutoff for monomer pairs. Interactions involving ions are purely repulsive apart from electrostatics and the ion–monomer solvation potential (2, 3),

$$U_{\text{solv}}(r) = -S \left[\left(\frac{\sigma}{r} \right)^4 - \left(\frac{\sigma}{r_c} \right)^4 \right], \quad r \leq r_c,$$

with $S = 4.33\epsilon$, $r_c = 5\sigma$ and $\epsilon = k_B T$. Each concentration is equilibrated at zero pressure and sampled at its mean volume. The two dielectric constants use the same interaction parameters apart from electrostatics. Four runs at each training concentration are held out, covering the different potential families, and three further runs are used for validation. Densities and internal force densities are recorded on 0.1σ bins. Profile uncertainties use eight time blocks. In the covariance-aware analysis, internal-force blocks are formed as $f_b^{\text{int}} = f_b^{\text{tot}} - f_b^{\text{ext}}$ before computing their standard error.

S2.2 Force matching. Let $\mathbf{R}$ denote all particle coordinates and $d\mathbf{R}$ the corresponding configurational measure. The normalized canonical distribution is

$$P(\mathbf{R}) = Z^{-1} \exp \left[-\frac{1}{k_B T} \left(U_{\text{int}}(\mathbf{R}) + \sum_{\alpha} \sum_{j \in \alpha} V_{\alpha}(\mathbf{r}_j) \right) \right],$$

and the one-body density is

$$n_{\alpha}(\mathbf{r}) = \left\langle \sum_{j \in \alpha} \delta(\mathbf{r} - \mathbf{r}_j) \right\rangle = \sum_{j \in \alpha} \int d\mathbf{R} P(\mathbf{R}) \delta(\mathbf{r} - \mathbf{r}_j).$$

Using $\nabla_{\mathbf{r}} \delta(\mathbf{r} - \mathbf{r}_j) = -\nabla_j \delta(\mathbf{r} - \mathbf{r}_j)$ and integrating by parts over $\mathbf{r}_j$, we obtain

$$\begin{aligned} \nabla n_{\alpha}(\mathbf{r}) &= \sum_{j \in \alpha} \int d\mathbf{R} P(\mathbf{R}) \nabla_{\mathbf{r}} \delta(\mathbf{r} - \mathbf{r}_j) \\ &= - \sum_{j \in \alpha} \int d\mathbf{R} P(\mathbf{R}) \nabla_j \delta(\mathbf{r} - \mathbf{r}_j) \\ &= \sum_{j \in \alpha} \int d\mathbf{R} \delta(\mathbf{r} - \mathbf{r}_j) \nabla_j P(\mathbf{R}). \end{aligned}$$

The boundary term vanishes for periodic boundaries (with the periodic delta function), or for sufficiently decaying distributions. For $j \in \alpha$,

$$\nabla_j P(\mathbf{R}) = -\frac{P(\mathbf{R})}{k_B T} [\nabla_j U_{\text{int}}(\mathbf{R}) + \nabla_j V_{\alpha}(\mathbf{r}_j)].$$

Substitution therefore gives

$$\begin{aligned} k_B T \nabla n_{\alpha}(\mathbf{r}) &= - \left\langle \sum_{j \in \alpha} \delta(\mathbf{r} - \mathbf{r}_j) \nabla_j U_{\text{int}} \right\rangle \\ &\quad - \left\langle \sum_{j \in \alpha} \delta(\mathbf{r} - \mathbf{r}_j) \nabla_j V_{\alpha}(\mathbf{r}_j) \right\rangle. \end{aligned}$$

Since the delta function fixes $\mathbf{r}_j = \mathbf{r}$,

$$\left\langle \sum_{j \in \alpha} \delta(\mathbf{r} - \mathbf{r}_j) \nabla_j V_{\alpha}(\mathbf{r}_j) \right\rangle = n_{\alpha}(\mathbf{r}) \nabla V_{\alpha}(\mathbf{r}).$$

Defining the internal force density by

$$\mathbf{f}_{\alpha}^{\text{int}}(\mathbf{r}) = - \left\langle \sum_{j \in \alpha} \delta(\mathbf{r} - \mathbf{r}_j) \nabla_j U_{\text{int}} \right\rangle,$$

we recover the local force balance (4, 5),

$$k_B T \nabla n_{\alpha}(\mathbf{r}) = \mathbf{f}_{\alpha}^{\text{int}}(\mathbf{r}) - n_{\alpha}(\mathbf{r}) \nabla V_{\alpha}(\mathbf{r}). \quad [\text{S9}]$$

Taking the gradient of the Euler–Lagrange equation and using Eq. S9 yields $\mathbf{f}_{\alpha}^{\text{int}} = -n_{\alpha} \nabla(e_{\alpha} \phi + \mu_{\alpha}^{\text{ex}})$, the force-matching relation of the main text.

The six Gaussian envelope widths are 0.250, 0.330, 0.435, 0.574, 0.758 and 1.000σ . The pointwise network receives $h_{m\alpha}/n_{\text{ref}}$, with $n_{\text{ref}} = 0.02\sigma^{-3}$, and its output contributes $n_{\text{ref}} \int \Phi_{\theta} d\mathbf{r}$ to the free energy. Other architecture parameters are given in the main text. The kernel range and activation were selected using only the validation profiles at $\varepsilon_r = 2$.

Training uses full batches and the AdamW schedule specified in the main text, with learning rate reduced to 10^{-5} . Validation is evaluated every 20 steps. Early stopping uses a patience of 1000 steps after at least 200 steps, and the best validation parameters are retained. Table S1 compares training with and without the Fourier term, using the same architecture and data splits. The Fourier term improves the long-wavelength amplitude and number response and leaves the profile errors unchanged.

Table S1. Loss ablation with and without the Fourier term weighted by $(k\sigma)^{-2}$. The amplitude ratio is the predicted over MD amplitude of the k_1 mode summed over the six long-wavelength training runs at $c = 0.04$. At this concentration, MD gives $\Gamma(k_1) = 4.38 \pm 0.15$ and 3.14 ± 0.56 at $\varepsilon_r = 7.5$ and 2, respectively. In the last two columns, values before and after the slash correspond to neutral ($\psi = 0$) and mixed neutral/charged ($V_N \neq 0, \psi \neq 0$) external potentials, respectively.

ε_r	Fourier term	Ratio at k_1	$\Gamma(k_1)$	Profile (%)	$k_1/2$ (%)	$k_1/4$ (%)
7.5	without	0.95	4.56	1.45	18 / 8	13 / 8
7.5	with	0.96	4.47	1.50	16 / 6	11 / 6
2	without	0.87	3.85	3.09	11 / 10	15 / 17
2	with	0.92	3.70	3.11	7 / 7	12 / 14

S2.3 Predictions and uncertainty. At fixed particle numbers, the predicted planar profiles satisfy

$$n_{\alpha}(z) = N_{\alpha} \frac{e^{-u_{\alpha}(z)}}{A \int_0^L e^{-u_{\alpha}(z')} dz'}, \quad u_{\alpha} = V_{\alpha} + e_{\alpha} \phi[\mathbf{n}] + \mu_{\alpha}^{\text{ex}}[\mathbf{n}], \quad [\text{S10}]$$

where A is the cross-sectional area. The chemical potentials enforcing particle numbers drop out. We solve for u_{α} using Anderson acceleration with history eight and mixing parameter 0.1 (6, 7). The charge residual is preconditioned by $k^2/(k^2 + \kappa_D^2)$ (8). Convergence requires the maximum density change divided by $\bar{n}$ to be below 10^{-6} . Profile errors are $\|n_{\alpha}^{\text{pred}} - n_{\alpha}^{\text{MD}}\|_2 / \|n_{\alpha}^{\text{MD}}\|_2$, averaged over runs.

The bulk kernel $W_{\alpha\beta}(k)$ follows by automatic differentiation of μ_{α}^{ex} at uniform density along a one-bin perturbation of species β , followed by Fourier transformation. Equation S7 gives $S(k)$. The neural functionals used for the main results have no negative eigenvalues of $S(k)^{-1}$ at the sampled concentrations and wavenumbers of a box eight times longer than the simulation box.

MD structure factors are obtained from zero-field runs. For each frame, we compute the microscopic Fourier densities

$$\tilde{n}_{\alpha}(\mathbf{k}, t) = \sum_{j \in \alpha} e^{-i\mathbf{k} \cdot \mathbf{r}_j(t)}, \quad \mathbf{k} = \frac{2\pi}{L} \mathbf{m}, \quad \mathbf{m} \in \mathbb{Z}^3.$$

For the homogeneous system at nonzero wavevectors, $\langle \tilde{n}_{\alpha}(\mathbf{k}, t) \rangle = 0$. The number structure factor is therefore

$$S_{NN}(k) = \frac{1}{V} \left\langle |\tilde{n}_{+}(\mathbf{k}, t) + \tilde{n}_{-}(\mathbf{k}, t)|^2 \right\rangle_{t, |\mathbf{k}|=k},$$

where $V = L^3$ and the average is over time and wavevector directions with the same magnitude.s

S3. Predictive accuracy and many-body correlations

S3.1 Predictions across concentrations. Table S2 collects the concentration-resolved force residuals and profile errors at both dielectric constants. Table S3 gives the corresponding bulk structure-factor errors and finite-wavenumber number response. The lowest concentration at strong coupling is the least accurate state for the number structure factor. The representative profiles and structure factors are shown in the main text.

Table S2. Predictions by concentration. Force residual χ^2 and relative profile error in percent (cation / anion), averaged over held-out runs. At $c = 0.05$ all 22 runs are untrained.

ε_r	c	NF χ^2	NF profile (%)	Pair χ^2	Pair profile (%)
7.5	0.01	0.559	3.5 / 2.7	1.82	5.7 / 8.6
7.5	0.02	0.574	1.8 / 1.4	3.25	4.9 / 7.4
7.5	0.04	0.573	1.2 / 0.8	2.12	2.7 / 3.2
7.5	0.05	0.606	1.0 / 0.7	1.51	2.0 / 1.8
7.5	0.06	0.615	0.9 / 0.6	0.82	1.5 / 1.0
7.5	0.08	0.837	1.4 / 0.8	2.94	5.2 / 1.9
2	0.01	0.693	9.2 / 8.9	6.28	14.9 / 15.7
2	0.02	0.618	2.8 / 2.6	20.50	14.9 / 15.8
2	0.04	2.559	1.6 / 2.3	20.21	6.6 / 7.1
2	0.05	0.748	1.0 / 0.9	4.56	2.1 / 3.1
2	0.06	0.865	1.1 / 0.9	2.42	1.0 / 2.9
2	0.08	1.154	1.3 / 0.5	19.88	3.7 / 5.2

Table S3. Bulk predictions at both dielectric constants. Structure-factor errors are relative rms deviations in percent over the MD wavevectors up to $k = 2.1\sigma^{-1}$, shown as neural functional / pair closure. $\Gamma = 2\bar{n}/S_{NN}(k_1)$ uses the MD uncertainty procedure of section S2.3.

ε_r	c	S_{ZZ} error	S_{NN} error	Γ MD	Γ NF	Γ pair
7.5	0.01	4.8 / 12.2	6.7 / 18.0	1.20 ± 0.06	1.34	1.96
7.5	0.02	4.1 / 12.6	3.6 / 16.1	1.97 ± 0.07	2.22	2.91
7.5	0.04	2.7 / 7.8	2.6 / 8.4	4.38 ± 0.15	4.47	4.76
7.5	0.05	3.6 / 5.2	2.8 / 4.3	5.42 ± 0.15	5.53	5.65
7.5	0.06	3.8 / 4.7	2.8 / 3.4	6.42 ± 0.26	6.73	6.51
7.5	0.08	3.9 / 13.0	2.4 / 13.5	9.26 ± 0.26	9.08	8.12
2	0.01	9.9 / 22.1	25.6 / 52.2	0.42 ± 0.07	0.31	1.84
2	0.02	4.5 / 31.6	5.3 / 38.3	0.98 ± 0.10	0.97	2.68
2	0.04	2.7 / 30.1	2.6 / 18.7	3.14 ± 0.56	3.70	4.32
2	0.05	3.5 / 23.8	2.7 / 8.1	4.41 ± 0.49	5.14	5.12
2	0.06	2.7 / 15.3	3.1 / 7.7	6.09 ± 0.91	6.77	5.88
2	0.08	2.7 / 10.4	2.8 / 42.1	11.05 ± 0.81	10.36	7.33

A pair closure can reproduce profiles at a single concentration but cannot adjust its bulk kernel with density. At $\varepsilon_r = 7.5$, fitting only $c = 0.04$ gives a held-out residual of 0.70 and cation/anion profile errors of 2.5/1.3%. Its held-out residual rises to 1.30 when $c = 0.01$ and 0.02 are added, to 1.55 after adding $c = 0.06$, and to 2.19 after adding $c = 0.08$. This comparison motivates the density-dependent correlations of the neural functional.

S3.2 Number response. For $\bar{n}_+ = \bar{n}_- = \bar{n}$, write the regular part of S^{-1} in the orthonormal number and charge basis, $\mathbf{e}_N = (1, 1)^\top/\sqrt{2}$ and $\mathbf{e}_Z = (1, -1)^\top/\sqrt{2}$. In the energy units of section S1,

$$A_{NN} = \bar{n}^{-1} + \frac{1}{2}(W_{++} + W_{--} + 2W_{+-}), \quad A_{ZZ} = \bar{n}^{-1} + \frac{1}{2}(W_{++} + W_{--} - 2W_{+-}), \quad A_{NZ} = \frac{1}{2}(W_{++} - W_{--}).$$

Inverting Eq. S7 gives

$$\Gamma(k) \equiv \frac{2\bar{n}}{S_{NN}(k)} = 1 + \frac{\bar{n}}{2} \sum_{\alpha\beta} W_{\alpha\beta}(k) - \frac{\bar{n}A_{NZ}^2}{A_{ZZ} + 8\pi l_B/k^2}. \quad [\text{S11}]$$

The last term accounts for coupling between number and charge fluctuations. It vanishes as $k \rightarrow 0$ and contributes 1–4% of Γ at k_1 in the reported models. Neglecting that correction yields the MD estimate used in the main text,

$$\mathbf{q}^\top W(k_1) \mathbf{q} \simeq \frac{2[\Gamma(k_1) - 1]}{\bar{n}}, \quad \mathbf{q} = (1, 1)^\top.$$

Multiplying by $k_B T$ restores energy units.

S4. Transfer and comparisons with other functionals

S4.1 Larger boxes and nonplanar densities. At $c = 0.04$ and both dielectric constants, we test transfer without retraining using the models trained on planar profiles. The tests include four aperiodic potentials absent from training, applied in a box twice the training length. They combine irregular Gaussian wells and barriers with Fourier modes at and between the training wavenumbers, with amplitudes within the trained range. Two contain no wavelength longer than the training box. The other two add a mode at $k_1/2$ with a peak-to-peak amplitude of $1 k_B T$, in the neutral part alone or in the neutral and charged parts. The neural functional reproduces the density profiles with relative L_2 errors of $1.4 - -1.6\%$ without the long mode and $0.8 - -1.6\%$ with it, compared with $2.4 - -6.2\%$ for the pair closure. The driven-mode amplitudes of the two halves of each MD run agree within 2.6 standard errors (median $0.6 - -1.0$).

Six potentials varying in x and y test transfer beyond planar geometry: an egg carton, neutral and charged Gaussian rods on a square lattice, a checkerboard with oblique wavevectors, and two aperiodic potentials. Aperiodic I combines six irregularly placed Gaussian wells and barriers and two oblique plane waves in the neutral part with three charged Gaussian wells. Aperiodic II combines three oblique plane waves and three Gaussian barriers with four irregularly placed charged wells. All amplitudes remain within the trained range. Single bins of the MD maps carry a sampling noise of 5–6%, so Table S4 reports the relative profile error on the Fourier modes that MD resolves, those with amplitudes above three standard errors, together with the MD noise level of the same quantity. On the aperiodic potentials, the neural functional reaches this noise level, with errors of 1.8–3.1% against noise levels of 1.7–3.0%, while the pair closure errs by 3.6–5.5% and Poisson–Boltzmann theory by 31–62%. On the periodic potentials, its errors are 0.6–3.7%, against 2.5–7.6% for the pair closure. The largest error occurs for the charged rods at $\varepsilon_r = 2$, 3.7% against 5.0% for the pair closure.

Table S4. Two-dimensional potentials at $c = 0.04$: relative profile errors (%) of the density maps predicted by the functionals trained on planar profiles. As for the planar profiles, the error is $\|n_\alpha^{\text{pred}} - n_\alpha^{\text{MD}}\|_2 / \|n_\alpha^{\text{MD}}\|_2$, averaged over cations and anions. It is evaluated on the Fourier modes of the maps that MD resolves, those with amplitudes above three standard errors (their number for both species is given), because single bins of the MD maps carry a sampling noise of 5–6%. The last column gives the MD noise level of the same quantity; an error at this level means agreement within the MD precision.

ε_r	potential	resolved modes	neural functional	pair closure	PB	MD noise
7.5	egg carton	32	1.0	3.5	40	1.0
	neutral rods	115	2.1	3.7	68	1.3
	charged rods	102	1.3	2.4	26	1.2
	checkerboard	54	1.2	2.9	42	1.1
	aperiodic I	327	3.1	3.6	61	3.0
	aperiodic II	218	3.0	4.1	33	2.6
2	egg carton	10	0.6	6.4	33	0.4
	neutral rods	114	2.2	5.4	70	1.1
	charged rods	96	3.7	5.0	15	1.0
	checkerboard	20	1.3	7.6	34	0.6
	aperiodic I	235	2.4	4.3	62	2.3
	aperiodic II	154	1.8	5.5	31	1.7

S4.2 Original formulations, adaptations and comparison protocol. Table 1 of the main text refers to the original formulations. The one-body networks of refs. (9, 10) apply a single network to a sliding window of a planar density profile on a fixed grid. They are therefore equivariant under translations by the grid spacing and can be applied to planar domains larger than their training boxes. Ref. (9) includes mirror symmetry only through data augmentation and verifies the thermal Noether sum rules and the symmetry of the pair direct correlation function numerically. Ref. (10) trains separate networks for cations and anions, so their cross derivatives need not coincide, and verifies the contact theorem instead. The convolutional free energy of ref. (11) applies average pooling after each layer to an input of fixed length, which breaks invariance under lattice translations and ties the model to the training box and grid. Ref. (12) builds invariance under the cubic point group O_h into finite-range environments on a fixed voxel grid and demonstrates transfer to larger cells and three-dimensional geometries at unchanged grid spacing. For the present functional, the guarantees hold for the continuous formulation (section S1); their finite-grid residuals are reported in section S4.3, and transfer to larger boxes and nonplanar densities in section S4.1.

Common protocol. Each adapted form replaces only the excess free energy F_{ex} of our functional or, for the one-body network, its excess chemical potentials μ_α^{ex} . Its input is the density profile on the 0.1σ bins of the MD data, continued periodically and divided by $n_{\text{ref}} = 0.02\sigma^{-3}$ ($0.4\sigma^{-3}$ for the Lennard-Jones fluid). For the ions, the Coulomb mean field F_{Coul} is kept analytic, as in our functional, so each network represents only the excess part. All forms are trained with the force-matching loss, optimizer, step budget and model selection of our functional, on the same splits (71 training runs at $\varepsilon_r = 7.5$ and 82 at $\varepsilon_r = 2$). The comparison therefore tests the architectures under one training protocol, not the published methods with their own data and training targets.

One-body network. Refs. (9, 10) fit the one-body direct correlation function $c_1 = -\mu^{\text{ex}}/k_B T$, obtained from grand-canonical simulations at known chemical potential, with a perceptron that acts on a window of the density profile. In ref. (9), the window spans $\pm 2.56\sigma$ at a resolution of 0.01σ (513 inputs) and the perceptron has three hidden layers of 512 units with smooth activations. Ref. (10) adds the long-range electrostatics through local molecular field theory. Our adaptation maps the cation and anion densities in a window of $\pm 3\sigma$ around z , 2×61 values, to both excess chemical potentials $\mu_{\theta,+}^{\text{ex}}(z)$ and $\mu_{\theta,-}^{\text{ex}}(z)$ with a single perceptron instead of one network per species. The perceptron has two hidden layers of 128 units with SiLU activations (32,514 parameters). The two outputs share hidden features, but $\partial\mu_{\theta,+}^{\text{ex}}/\partial n_-$ and $\partial\mu_{\theta,-}^{\text{ex}}/\partial n_+$ are not constrained to agree. The output enters force matching only through the force density $-n_\alpha \partial_z(e_\alpha \phi + \mu_{\theta,\alpha}^{\text{ex}})$, so canonical data without chemical-potential labels suffice. A window of $\pm 6\sigma$ (47,874 parameters) gives a validation residual 4% and 40% larger at $\varepsilon_r = 7.5$ and 2, and about ten times larger for the Lennard-Jones fluid.

Lattice free energy. Ref. (12) writes the excess free energy of a one-component fluid on a cubic grid of spacing $\Delta L = 0.5\sigma$ and voxel volume $\Delta V = \Delta L^3$ as $F_{\text{ex}} = \Delta V \sum_g \rho_g [a_{\text{loc}}(\rho_g) + a_{\text{CACE}}(\rho_g, \mathbf{B}_g)]$. The features $\mathbf{B}_g$ are built from the Cartesian moments $A_1(g) = \sum_{\mathbf{q}} \rho_{g+\mathbf{q}} q_x^{l_x} q_y^{l_y} q_z^{l_z}$ with $|\mathbf{l}| \leq 3$, summed with unit weight over the 122 nonzero integer offsets with $|\mathbf{q}| \leq 3$. Products of up to two moments, averaged over the 48 elements of O_h , give 15 invariants. Both read-outs have hidden layers of 32 and 16 SiLU units and also take the temperature (1,762 parameters in total). We form the moments separately for cations and anions. The first-order invariants of each species and the second-order invariants within and across species, averaged over O_h and with vanishing and duplicate forms removed, give 47 invariants (15 for one species, as in the original). Both read-outs have one output per species, and $F_{\text{ex}} = A \Delta z \sum_i \sum_\alpha \bar{n}_\alpha(z_i) a_\alpha(z_i)$, where A is the cross-sectional area and Δz the bin width. The voxel density $\bar{n}_\alpha(z_i)$ is the average of the profile over the five bins (0.5σ) centred on bin i . For a planar density, the lateral sums in the moments reduce to fixed weights, $A_1^\alpha(z_i) = \sum_{q_z} C_1(q_z) \bar{n}_\alpha(z_i + q_z \Delta L)$ with $C_1(q_z) = \sum_{q_x, q_y} q_x^{l_x} q_y^{l_y} q_z^{l_z}$ over the stencil, which are evaluated exactly; six of the 47 invariants (two of the 15) then vanish identically. Every bin is a voxel centre, so F_{ex} averages the lattice free energy over the registrations of the lattice on the bin grid and is invariant under translations by one bin. Densities are divided by n_{ref} , each invariant by its root-mean-square value over the training profiles, and temperature is not an input. The read-out a_{CACE} is widened to two hidden layers of 128 units, the width of our read-out, while a_{loc} keeps 32 and 16 (23,828 parameters). The excess chemical potentials $\mu_\alpha^{\text{ex}} = \delta F_{\text{ex}} / \delta n_\alpha$ follow by automatic differentiation and are trained by force matching. We keep the stencil radius of the original, 1.5σ . A radius of 3σ , with the same invariants and number of

parameters, raises the validation residual sevenfold at $\varepsilon_r = 7.5$ and 27-fold for the Lennard-Jones fluid. At $\varepsilon_r = 2$, it lowers the validation residual by about a quarter and the mean held-out profile error from 7.2% to 5.2%, but the Euler–Lagrange solution fails for 15 instead of 10 of the 42 test runs.

Convolutional free energy. Ref. (11) represents the excess free energy of a one-component fluid on 320 grid points ($\Delta z = \sigma/32$) by six periodic dilated convolutions (kernel size 3, dilation 2, channels 16, 16, 32, 32, 64, 64), each followed by average pooling over two points, and a scalar output. Its stated size of about 24,400 parameters implies a linear read-out of the flattened final feature map, 64 channels at five positions; the activation is not specified. The network is trained by pair-correlation matching: rows of its Hessian are fitted to the laterally integrated direct correlation functions of bulk grand-canonical simulations, together with an offset term for the first derivative. We keep the convolution and pooling stack. The input has two channels, n_+/n_{ref} and n_-/n_{ref} (one for the Lennard-Jones fluid), padding is periodic and the activations are softplus. Pooling drops an odd last point, so six poolings leave three positions for any grid of 192 to 255 bins, which covers all boxes here (243 to 250 bins for the ions, 240 for the Lennard-Jones fluid). A linear layer maps the 3×64 final features to F_{ex} (24,321 parameters). The network is trained by force matching on the inhomogeneous profiles instead of by pair-correlation matching on bulk data.

Lennard-Jones fluid. The truncated and shifted fluid has $r_c = 2.5\sigma$, $T = 1.5$ and $L = 24\sigma$. It has a single species and no Coulomb term, so each form takes one density; the forms then have 18,861 (ours), 24,577 (one-body network), 19,426 (lattice free energy) and 24,273 (convolutional free energy) parameters. The planar potentials use the same families as the ion data, 22 per density. One run each at 0.5 and $0.6\sigma^{-3}$ violates the force balance $k_B T \partial_z n = f^{\text{ext}} + f^{\text{int}}$ beyond its sampling error and is discarded. At the trained densities 0.1, 0.2, 0.4, 0.6 and $0.7\sigma^{-3}$, four runs per density are held out and three are used for validation, which leaves 74 training runs; the 21 runs at $0.5\sigma^{-3}$ are never trained. This comparison tests the adapted implementations on a simple fluid, but does not isolate the cause of their different performance on the ions.

S4.3 Data efficiency and structural consistency. Table S5 reports nested, stratified fractions of the ion training data, with validation and test runs fixed. The comparison in the main text uses these same fits.

Table S5. Data efficiency on the ions. Mean relative profile errors (%) of the held-out runs / of the runs at the untrained concentration $c = 0.05$, for the training fractions shown. Means exclude Euler–Lagrange solutions that do not converge or collapse onto a density spike (error above 50%). The one-body window is $\pm 3\sigma$ and the lattice stencil radius is 1.5σ .

ε_r	Training fraction	Neural functional	One-body network	Lattice free energy
7.5	0.25	1.55 / 0.92	3.00 / 2.18	4.53 / 3.98
7.5	0.5	1.48 / 0.84	2.09 / 1.27	3.72 / 3.32
7.5	1	1.50 / 0.85	1.75 / 1.01	2.97 / 2.06
2	0.25	3.56 / 1.28	5.55 / 1.96	6.31 / 5.12
2	0.5	3.45 / 1.02	4.95 / 1.24	6.00 / 2.54
2	1	3.11 / 0.98	3.41 / 0.85	7.16 / 2.72

Structural errors in Table S6 are evaluated on the MD profiles. The relative net internal force is $|\sum_{\alpha} \int f_{\alpha}^{\text{int}} dz| / \sum_{\alpha} \int |f_{\alpha}^{\text{int}}| dz$, including Coulomb forces. Integrability is tested by $\|J - J^T\| / \|J\|$ for $J = \partial \mu^{\text{ex}} / \partial n$. Reflection error is $\|\mu[Rn] - R\mu[n]\| / \|\mu - \langle \mu \rangle\|$, with $R: z \mapsto -z$. Path dependence is $|\oint \mu^{\text{ex}} \cdot dn| / \sum |\text{legs}|$ on closed paths $\bar{n} \rightarrow n_1 \rightarrow n_2 \rightarrow \bar{n}$ through held-out profiles. The continuum identities of section S1 and their finite-grid residuals are distinct from predictive accuracy.

Grid independence is tested by evaluating each form, with unchanged parameters, on the test profiles resampled by band-limited interpolation from the 0.1σ bins to grids of 0.05σ and 0.2σ . The rms change of μ_{θ}^{ex} relative to its variation over the box, averaged over the 42 test runs of each ε_r , is below 10^{-9} on the finer and 3×10^{-6} on the coarser grid for the neural functional (maxima 10^{-8} and 5×10^{-5}), and below 10^{-13} and 10^{-6} for the pair closure. The one-body network changes by 65–79% and the lattice free energy by 62–77%, because their window and voxel are fixed numbers of bins; the convolutional free energy cannot be evaluated, because its read-out fixes the number of grid points.

Table S6. Structural errors for the selected implementations. Values are maxima over runs and state points. Parentheses give arithmetic means over runs where available.

Form	ε_r	Net force	Jacobian asymmetry	Reflection	Path dependence
Neural functional	7.5	10^{-7} (10^{-9})	2×10^{-16} (2×10^{-16})	10^{-14}	2×10^{-14}
Neural functional	2	2×10^{-6} (10^{-8})	3×10^{-16} (2×10^{-16})	2×10^{-14}	2×10^{-12}
One-body network	7.5	0.14 (0.008)	0.55 (0.31)	0.38	0.008
One-body network	2	0.08 (0.005)	0.46 (0.25)	0.16	0.009
Lattice free energy	7.5	7×10^{-4} (7×10^{-6})	2×10^{-16} (2×10^{-16})	4×10^{-15}	2×10^{-10}
Lattice free energy	2	2×10^{-5} (3×10^{-7})	2×10^{-16} (10^{-16})	10^{-14}	7×10^{-8}

S4.4 Convolutional free-energy baseline. Table S7 compares the convolutional free energy of section S4.2 with our functional, both trained by force matching on the same data. Its scalar output preserves integrability, but pooling and the read-out break translation and reflection invariance, so its bulk Hessian is not diagonal in k . Its structure factors use the Fourier diagonal of the Hessian at uniform density, averaged over eight phase-aligned rows. The convolutional form has larger force and profile errors, and few of its Euler–Lagrange solutions converge.

Table S7. Force-matching comparison of the neural and convolutional free energies. Structure-factor errors are the largest relative rms errors across $c = 0.02$ – 0.08 (S_{ZZ} / S_{NN}).

ε_r	Form	Held-out χ^2	Profile (%)	EL converged	$S(k)$ error (%)
7.5	neural functional	0.63	1.50	42/42	4.1 / 3.6
7.5	convolutional	83.4	13.8	5/42	1171 / 160
2	neural functional	1.18	3.11	42/42	4.5 / 5.3
2	convolutional	97.0	15.1	0/42	48 / 30

Convolutional profile errors include Euler–Lagrange iterations that did not converge. EL counts cover all 42 test runs.

References

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